



11940 - Toward the First Electromagnetic Detection of an Isolated Stellar-Mass Black Hole

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRCAM				
	1		NIRCam Imaging	(1) OGLE-2011-BLG-0462

ABSTRACT

Stellar-mass isolated black holes (IsoBHs) represent a long-predicted but largely hidden population of compact objects drifting through the Galaxy. By accreting material from the ISM, they are expected to produce faint electromagnetic emission spanning radio to gamma-ray wavelengths. Detecting such emission would provide insights into the accretion physics and potentially provide a new window to reveal the hidden population of dormant black holes in our Galaxy.

OGLE-2011-BLG-0462 is the first and only confirmed IsoBH, discovered through astrometric microlensing. With a mass of 7.15 solar mass and a distance of 1.5 kpc in the Sagittarius arm, OGLE-2011-BLG-0462 offers a unique opportunity to directly probe accretion in an isolated environment. Recent Chandra observations yielded no X-ray detection, limiting the range of possible models and motivating deeper exploration at longer wavelengths.

Theoretical models suggest that JWST/NIRCam observations provide the best chance to detect the accretion emission. For plausible ISM densities and accretion rates, a 3.5-hour exposure should lead to a detection of the accretion signal in both the SW and LW channels—potentially enabling the first-ever direct detection of radiation from an IsoBH.

Such a detection would mark a major advance in black hole astrophysics, offering clear insights into the physics of accretion and the demographics of Galactic IsoBHs. With estimates suggesting ~ 35 IsoBHs within 50 pc of the Sun, JWST observations of OGLE-2011-BLG-0462 could establish a blueprint for discovering this elusive population.

OBSERVING DESCRIPTION

Our goal is to detect the faint accretion flux near a much brighter source, which requires both high spatial resolution and high sensitivity. We will use the F150W2 and F322W2 filters simultaneously, as they offer the highest sensitivity in the SW and LW channels, respectively. To minimize saturation from the nearby bright star, we will use the BRIGHT2 readout pattern.

The BH (serving as the lens) has a proper motion of 7.35 mas/yr relative to the source (Sahu et al. 2025). With JWST observations scheduled for late 2025—about 14 years after the microlensing event—the lens will be separated by 103 mas from the bright source. We estimate that the nearby bright source will be approximately 6 to 9 magnitudes brighter than the accretion flux from OB-11-462. This large contrast necessitates accurate PSF subtraction of the nearby star to enable detection of the faint accretion signal. To achieve this, we will observe at nine different dither positions, and use our expertise to construct a well-sampled PSF using reference stars in the field. Importantly, JWST/NIRCam observations are not affected by charge bleeding, making it an ideal instrument for this study.

The SW channel has a pixel scale of 31 mas/pixel and offers a spatial resolution of approximately 80 mas with our chosen F150W2 filter. This resolution should allow the BH to be detected as a distinct source in the SW channel following PSF subtraction of the nearby star. The LW channel, with 63 mas/pixel and correspondingly lower spatial resolution, is nonetheless valuable for detecting the accretion flux at longer wavelengths. These LW observations will help constrain the spectral energy distribution of the accretion spectrum.

Proposal 11940 - Targets - Toward the First Electromagnetic Detection of an Isolated Stellar-Mass Black Hole

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	OGLE-2011-BLG-0462	RA: 17 51 40.2080 (267.9175333d) Dec: -29 53 26.50 (-29.89069d) Equinox: J2000	Epoch of Position: 2000	
Comments: Category=Star Description=[Red giants]					

Proposal 11940 - Observation 1 - Toward the First Electromagnetic Detection of an Isolated Stellar-Mass Black Hole

Observation	Proposal 11940, Observation 1									Fri Mar 13 22:05:34 GMT 2026	
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Data Excess over lower threshold										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	OGLE-2011-BLG-0462	RA: 17 51 40.2080 (267.9175333d)			Epoch of Position: 2000					
			Dec: -29 53 26.50 (-29.89069d)								
			Equinox: J2000								
Template	Comments: Category=Star Description=[Red giants]										
	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRASCA		9		STANDARD		16" (MEDIUM)		1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W2	F322W2	BRIGHT2	8	8	72	9	13045.176		